SEQUENCES, SERIES
Examination questions

1 An arithmetic series has a as its first term, d as the common difference.

(a)(i) Write down the formula for T_n, the nth term.

(ii) Use this formula to prove that S_n, the sum of the first n terms is given by

$$S_n = \frac{n}{2}[2a + (n-1)d].$$

(b) Twice the sum of the first five terms of an arithmetic series is equal to five times the seventh term. Also the sum of the fourth and fifth terms is 22.

Find the sum of the first fifteen terms.

2 The first and last terms of an arithmetic series are 10 and 66, respectively, and the sum of the series is 570.

Find: (a) the number of terms in the series

(b) the common difference

3 The first three terms of an arithmetic series are 15, 12 and 9.

(i) Find the twentieth term of this series.

(ii) Find the sum of the first twenty terms.

4 Given that $S_n = \frac{n}{2}[2a + (n-1)d]$, find S_n when $n = 97$, $a = 3$ and $d = 1{\cdot}5$.

5 A telecommunications company sells 1800 mobile phones in the first month of operating. The owners plan to increase their sales by 200 mobile phones each month.

How many mobile phones do they plan to sell

(i) in the last month of the third year of operation?

(ii) over the entire three-year period?

6 The sum of the first n terms of a certain arithmetic series is given by

$$S_n = \frac{n(3n+7)}{2}.$$

(i) Calculate S_1 and S_2.

(ii) Find the first three terms of the series.

(iii) Find an expression of T_n, the nth term.

7 The ninth term of an arithmetic sequence is 45 and the sixteenth term is 73.

(i) Find the values of the first term and the common difference.

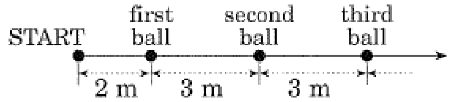
(ii) Find the sum of the first 25 terms.

8 A worker at a factory is stacking cylindrical-shaped hollow pipes which are stacked in layers. Each layer contains one pipe less than the layer below it. There are five pipes in the topmost layer, six in the next layer, and so on. There are n pipes in the stack.

(i) Determine the number of pipes in the bottom layer.

(ii) Show that there are $\frac{n(n+9)}{2}$ pipes in the stack.

9



At a social club's annual picnic, competitors take part in the following event:

> A number of tennis balls are placed in a straight line. The first ball is 2 metres from the start line, and there is a 3 metre distance between each of the remaining consecutive balls. There are n tennis balls placed out in the line.

Each competitor runs from the start line, collects the nearest ball and runs back with it to the start line, where it is deposited in a box. The competitor runs back to collect the next ball, returning with it to the start line. The same pattern continues until the competitor has collected all n balls.

(i) How far does the competitor run to collect and deposit the kth ball?

(ii) How far does the competitor run to collect and deposit all n balls?

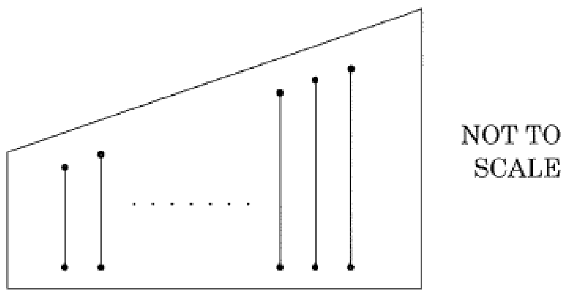
(iii) The competitor runs 1220 metres to collect and deposit all the tennis balls. How many tennis balls are there?

10 The numbers 8, 16, 24, … form a sequence of positive multiples of 8.

(i) What is the largest multiple of 8 less than 1000?

(ii) What is the sum of the positive multiples of 8 which are less than 1000?

11



A piece of plyboard has many slits cut into it, as in the diagram above.

Each consecutive slit has a constant difference in length, so that the lengths of the slits form the terms of an arithmetic sequence.
The shortest slit is 30 cm long and the longest slit is 50 cm. The sum of the lengths of all of the slits is 1240 cm.

(i) Find the number of slits.
(ii) Find the constant difference in length between consecutive slits.

12 If a number is added to each of 2, 6 and 13, the resulting set of numbers form a geometric progression. Find the number.

13 A geometric series has a third term of 32 and the ratio of the tenth term to the ninth term is 4.

(i) Find the common ratio r.
(ii) What is the first term a?
(iii) Find the sum of the first 12 terms.

14 (i) It is known that the geometric series $a + ar + ar^2 + \cdots$ has a limiting sum. Write down the limiting sum for these values of r.

(ii) Find a geometric series with common ratio $\frac{1}{p}$ and which has $\frac{1}{1-p}$ as its limiting sum.

15 Express $0{\cdot}\dot{3}\dot{1}$ as a simple fraction.

16 A sum of \$20 000 is invested, earning a rate of 8·5% p.a., compounded annually.
Find the amount that the investment has become at the end of 7 years, immediately after the final interest has been paid. Give your answer to the nearest dollar.

17 Maria invests \$2000 at 9% per year compound interest, compounded monthly. Calculate the total interest earned by the investment after 6 years.

18 Ben decides to invest \$800 at the start of each year into a superannuation fund. The interest is compounded at a rate of 10% per annum on the amounts invested. The first \$800 is to be invested at the beginning of 2001 and the last is to be invested at the beginning of 2030.
Calculate, to the nearest dollar,

(i) the amount to which the 2001 investment will have grown by the beginning of 2031;
(ii) the amount to which the total investment will have grown by the beginning of 2031.

19 A scientific researcher was studying the population growth of a species of insect.
He knew the population grew at a rate of 1%, compounding daily.
The initial insect population was 1000.

(i) How many insects would be in the colony after 100 days? (You may assume the mortality rate is zero over this period of time.)
(ii) Suppose he had introduced 1000 insects at the beginning of each day, with the same rate of population growth compounding daily.
How many insects would be in the colony after 200 days just before the next 1000 insects would have been added?
Give your answer to the nearest thousand.
(Once again, assume a zero mortality rate over this time.)

20 A small-business owner borrows \$50 000 to expand his business. The interest is calculated monthly at a rate of 1% per month, and is compounded each month.
The loan is to be repaid in two equal annual instalments of \$$M$ at the end of the first and second years.

(i) How much does the owner owe at the end of the first month?
(ii) Write an expression involving M for the total amount owed after 12 months, just after the first instalment of \$$M$ has been paid.
(iii) Find an expression for the amount owed at the end of the second year and show that
$$M = \frac{50\,000 \times (1{\cdot}01)^{24}}{1{\cdot}01^{12} + 1}.$$
(iv) What is the total interest over the two-year period?

21 Justin invested \$10 000 into an investment account on 1 January 2000. The investment earned interest at a fixed rate of 8% per annum, compounding each year.

(i) How much would be in the investment account after the interest payment made on the 1 January 2010? You may assume Justin made no additional payments during this period.
(ii) Suppose that Justin had in fact added \$1000 to his account on 1 January each year, beginning on 1 January 2001. How much would be in the account on 1 January 2010 after the payment of interest, including his deposit? The same rate of interest applies as in part (i).
(iii) Justin's girlfriend Gina invested \$10 000 in an account with another finance institution on 1 January 2000, making no further deposits. On 1 January 2010, the balance of Gina's account was \$35 478. What was the annual rate of compound interest that Gina received on her investment?

22 Glenys invests \$60 000 in an account which earns interest at a compound rate of 7% per annum. She wants to make a withdrawal of $\$W$ at the end of each year for part of her living expenses. The withdrawal is made immediately after that year's interest has been paid.

She intends to do this for 20 years, so that at the end of this time her account will have a zero balance.

(i) How much money will be in her account immediately after she has made the first withdrawal?

(ii) Write an expression in terms of W for the amount of money remaining in her account just before her final withdrawal.

(iii) Calculate the value of P which leaves the account with a zero balance just after the 20th withdrawal.

(iv) Assume Glenys wishes to be able to withdraw \$7500 per annum for the 20 years. Estimate, to the nearest per cent, the interest rate she would need to earn on her investment.

SEQUENCES, SERIES
Worked solutions to examination questions

1 (a) (i) $T_n = a + (n-1)d$

(ii)
$$\begin{aligned}
S_n &= \quad a \quad + \quad (a+d) \quad + \quad (a+2d) \quad + \cdots + \quad a+(n-1)d \\
S_n &= a+(n-1)d + a+(n-2)d + a+(n-3)d + \cdots + \quad a \\
2S_n &= \underbrace{[2a+(n-1)d]+[2a+(n-1)d]+[2a+(n-1)d]+\cdots+[2a+(n-1)d]}_{n \text{ terms}}
\end{aligned}$$

Write the original line in reverse, and add both lines.

$\therefore \quad 2S_n = n[2a+(n-1)d]$

$$S_n = \frac{n}{2}[2a+(n-1)d]$$

(b) $2S_5 = 5T_7$ $\qquad T_4 + T_5 = 22$

$S_5 = \frac{5}{2}[2a+4d]$ $\qquad (a+3d)+(a+4d) = 22$

$T_7 = a+6d$ $\qquad \therefore \quad 2a+7d = 22$ —①

$$\begin{aligned}
\therefore \quad 2\times\frac{5}{2}[2a+4d] &= 5(a+6d) \\
5[2a+4d] &= 5a+30d \\
10a+20d &= 5a+30d \\
\therefore \quad 5a-10d &= 0 \\
\therefore \quad a-2d &= 0 \quad \text{—②}
\end{aligned}$$

Solving ① and ② simultaneously, we have

$$\begin{aligned}
2a+7d &= 22 \quad \text{—①} \\
2a-4d &= 0 \quad \text{—②} \times 2 \\
\therefore \quad 11d &= 22 \\
d &= 2
\end{aligned}$$

Substituting in ①:
$$\begin{aligned}
2a+7(2) &= 22 \\
2a+14 &= 22 \\
2a &= 8 \\
a &= 4
\end{aligned}$$

$$\begin{aligned}
S_{15} &= \frac{15}{2}[2(4)+(15-1)2] \\
&= \frac{15}{2}(8+28) \\
&= \frac{15}{2}\times 36 \\
&= 15\times 18 \\
&= 270
\end{aligned}$$

2 (a) $T_1 = a = 10$

$T_n = a+(n-1)d$ $\qquad \therefore \quad 10+(n-1)d = 66$

$\qquad (n-1)d = 56$

$$\begin{aligned}
S_n = \frac{n}{2}[2a+(n-1)d] &= 570 \\
\therefore \quad \frac{n}{2}[2(10)+56] &= 570 \\
\frac{n}{2}(76) &= 570 \\
38n &= 570 \\
\therefore \quad n &= 15
\end{aligned}$$

(b)
$$\begin{aligned}
(n-1)d &= 56 \\
\therefore \quad (15-1)d &= 56 \\
14d &= 56 \\
d &= 4
\end{aligned}$$

3 15, 12, 9, …

(i) Since the sequence forms an AP, then $a = 15,\ d = -3$

$$\begin{aligned} T_n &= a+(n-1)d \\ T_{20} &= 15+19(-3) \\ &= 15-57 \\ &= -42 \end{aligned}$$

(ii)
$$\begin{aligned} S_n &= \frac{n}{2}(a+\ell) \\ S_{20} &= \frac{20}{2}[15+(-42)] \\ &= 10(-27) \\ &= -270 \end{aligned}$$

4
$$\begin{aligned} S_n &= \frac{n}{2}[2a+(n-1)d] \\ S_{97} &= \frac{97}{2}[2(3)+96(1{\cdot}5)] \\ &= \frac{97}{2}(6+144) \\ &= \frac{97}{2}\times 150 \\ &= 7275 \end{aligned}$$

5 The information represents an AP, with $a = 1800,\ d = 200,\ n = 36$

$\boxed{n = 3 \text{ years} = 36 \text{ months}}$

(i)
$$\begin{aligned} T_n &= a+(n-1)d \\ T_{36} &= 1800+35(200) \\ &= 1800+7000 \\ &= 8800 \end{aligned}$$

(ii)
$$\begin{aligned} S_n &= \frac{n}{2}[2a+(n-1)d] \\ S_{36} &= \frac{36}{2}[2(1800)+35(200)] \\ &= 18(3600+7000) \\ &= 18(10\,600) \\ &= 190\,800 \end{aligned}$$

6 $S_n = \dfrac{n(3n+7)}{2}$

(i)
$$S_1 = \frac{1(3+7)}{2} = \frac{10}{2} = 5$$
$$S_2 = \frac{2(6+7)}{2} = 13$$

(ii)
$$\begin{aligned} S_1 &= T_1 = a = 5 \\ S_2 &= T_1+T_2 = 5+T_2 = 13 \quad \text{from (i)} \\ \therefore\quad & T_2 = 8 \\ T_2 &= a+d = 5+d = 8 \quad \therefore\ d = 3 \\ T_3 &= a+2d = 5+2(3) = 5+6 = 11 \end{aligned}$$

i.e. $T_1 = 5;\ T_2 = 8;\ T_3 = 11$

(iii)
$$\begin{aligned} T_n &= a+(n-1)d \\ &= 5+(n-1)3 \\ &= 5+3n-3 \end{aligned}$$

i.e. $T_n = 2+3n$

$\boxed{T_n = a + (n-1)d}$

7 (i)
$$T_9 = 45 = a + 8d \quad —①$$
$$T_{16} = 73 = a + 15d \quad —②$$
② − ①: $28 = 7d$
$$d = 4$$
Substituting in ①:
$$45 = a + 8(4)$$
$$= a + 32$$
$$a = 13$$
i.e. $a = 13,\ d = 4$

(ii) The series is $13 + 17 + 21 + \cdots$
$$S_n = \frac{n}{2}[2a + (n-1)d]$$
$$S_{25} = \frac{25}{2}[2(13) + (24)4]$$
$$= \frac{25}{2}(26 + 96)$$
$$= \frac{25}{2}(122)$$
$$= 1525$$

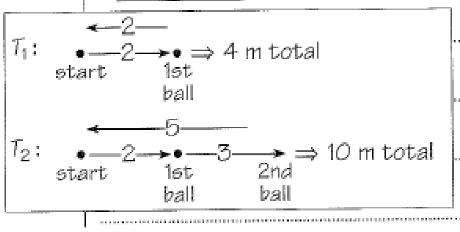
What type of sequence is formed?

8 (i) The stack is equivalent to an AP with T_1 as the top layer, T_2 the layer below, and so on

$\therefore$ the series is $5, 6, 7, \ldots, T_n$
$$T_n = a + (n-1)d$$
$$= 5 + (n-1)\,1$$
$$= 5 + n - 1$$
$$= 4 + n$$
$\therefore$ there are $(4 + n)$ pipes in the bottom layer.

(ii) The number of pipes is equivalent to finding S_n
$$S_n = \frac{n}{2}(a + \ell)$$
$$= \frac{n}{2}[5 + (4 + n)]$$
$$= \frac{n(n+9)}{2}$$
$\therefore$ total number of pipes is $\dfrac{n(n+9)}{2}$.

List the distances run to each ball as a sequence. What type of sequence is being formed?



9
$$T_1 = 4 = a \quad —①$$
$$T_2 = 10 = a + d \quad —②$$
$$T_3 = 16 = a + 2d$$
From ① and ②: $a = 4,\ d = 6$

(i)
$$T_n = a + (n-1)d$$
$$= 4 + (n-1)6$$
$$= 4 + 6n - 6$$
$$= 6n - 2$$
$$\therefore \quad T_k = 6k - 2$$
$\therefore$ the competitor runs $(6k - 2)$ metres to collect and deposit the kth ball.

(ii) Total distance is given by S_n.
$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}[2(4) + (n-1)6]$$
$$= \frac{n}{2}(8 + 6n - 6)$$
$$= \frac{n}{2}(2 + 6n)$$
$$= n(1 + 3n)$$

(iii) $S_n = 1220 = n(3n+1)$

$\therefore \quad 3n^2+3n-1220 = 0$

$(3n+61)(n-20) = 0$

$\therefore \ n = 20, -\dfrac{61}{3}$

Rejecting $n = -\dfrac{61}{3}$, $n > 0$, then $n = 20$

$\therefore$ there are 20 tennis balls.

10 (i) For the sequence 8, 16, 24, …, $8n$

$T_n = 8n < 1000$

$\therefore \quad n < \dfrac{1000}{8} = 125$

i.e. $\quad n = 124$

$\therefore$ the largest multiple of 8 less than 1000 $= 124 \times 8 = 992$

Note the multiple of 8 is less than 1000.

(ii) $S_n = \dfrac{n}{2}(a+\ell)$

$\therefore \quad S_{124} = \dfrac{124}{2}(8+992)$

$= 62(1000)$

$= 62\,000$

11 (i) The sequence is 30, …, 50

$T_1 = 30 = a$

$T_n = 50 = \ell$

$S_n = 1240 = \dfrac{n}{2}(a+\ell) \quad \therefore \quad 1240 = \dfrac{n}{2}(30+50) = 40n$

$n = \dfrac{1240}{40} = 31$

$\therefore$ the number of slits is 31

(ii) $T_1 = a = 30;\ n = 31;\ T_n = 50$

$T_n = a+(n-1)d$

$50 = 30+(31-1)d$

$\therefore \quad 20 = 30d$

i.e. $\quad d = \dfrac{20}{30} = \dfrac{2}{3}$ cm

$\therefore$ the difference in length of each slit is $\dfrac{2}{3}$ cm

12 Let the number be x

$\therefore$ the sequence is $2+x,\ 6+x,\ 13+x$

Since the terms form a GP, the common ratio $r = \dfrac{13+x}{6+x} = \dfrac{6+x}{2+x}$

$\therefore \quad (13+x)(2+x) = (6+x)(6+x)$

$26+15x+x^2 = 36+12x+x^2$

$\therefore \quad 3x = 10$

$x = \dfrac{10}{3} = 3\frac{1}{3}$

13 (i) $T_3 = ar^2 = 32$

$\dfrac{T_{10}}{T_9} = \dfrac{ar^9}{ar^8} = 4 \qquad \therefore \quad r = 4$

$T_n = ar^{n-1}$

Express $\dfrac{T_{10}}{T_9}$ in terms of a, r

(ii) $T_3 = a\left(4^2\right) = 32 \qquad \therefore \ 16a = 32$

$a = 2$

(iii) $S_n = \dfrac{a(r^n - 1)}{r - 1}$ $\quad\therefore\quad S_{12} = \dfrac{2(4^{12} - 1)}{4 - 1}$

$$= \frac{2(16\,777\,215)}{3}$$

$$= 11\,184\,810$$

14 (i) The limiting sum $S_\infty = \dfrac{a}{1-r}$ for $-1 < r < 1$

(ii) $S_\infty = \dfrac{a}{1-r}$ with $r = \dfrac{1}{p}$

$$\therefore \quad \frac{1}{1-p} = \frac{a}{1-\frac{1}{p}} = \frac{a}{\frac{p-1}{p}}$$

$$\therefore \quad \frac{1}{1-p} \times \frac{p-1}{p} = a$$

$$\therefore \quad a = \frac{1}{1-p} \times \frac{-(1-p)}{p} = -\frac{1}{p}$$

Recognise $(p-1)$ as the negative of $(1-p)$.

$\therefore$ sequence is $-\dfrac{1}{p}, -\dfrac{1}{p}\left(\dfrac{1}{p}\right), -\dfrac{1}{p}\left(\dfrac{1}{p}\right)^2, \ldots$

i.e. $-\dfrac{1}{p}, -\dfrac{1}{p^2}, -\dfrac{1}{p^3}, \ldots$

$\therefore$ the series is $-\dfrac{1}{p} - \dfrac{1}{p^2} - \dfrac{1}{p^3} - \cdots$

15 $0{\cdot}0\dot{3}\dot{1} = 0{\cdot}313131\ldots$

$$= \frac{31}{100} + \frac{31}{10\,000} + \cdots$$

This forms a limiting sum with $a = \dfrac{31}{100}$, $r = \dfrac{1}{100}$:

$$S_\infty = \frac{a}{1-r} = \frac{\frac{31}{100}}{1-\frac{1}{100}} = \frac{\frac{31}{100}}{\frac{99}{100}} = \frac{31}{99}$$

$$\frac{\frac{a}{b}}{\frac{c}{b}} = \frac{a}{c}$$

Check your answer by converting your result back to a decimal.

16 $A = P\left(1+\dfrac{r}{100}\right)^n = 20\,000\left(1+\dfrac{8{\cdot}5}{100}\right)^7$

$$= 20\,000(1{\cdot}085)^7$$

$$= \$35\,402{\cdot}84$$

$$\approx \$35\,403 \text{ to the nearest dollar}$$

16 Compound interest formula

17 $A = P\left(1+\dfrac{r}{100}\right)^n$; $r = 9\%$ p.a. $= \dfrac{9}{12}\%$ p.m. $= 0{\cdot}75\%$ p.m.

$n = 6$ years $= 72$ months

17 Be careful of the time units involved.

$$= 2000\left(1+\frac{0{\cdot}75}{100}\right)^{72}$$

$$= 2000(1{\cdot}0075)^{72}$$

$$= \$3425{\cdot}105$$

$$\approx \$3425{\cdot}11$$

Interest $= A - P$

$$= \$3425{\cdot}11 - \$2000$$

$$= \$1425{\cdot}11$$

18 (i) Amount (after 30 years):

$$\begin{aligned} A &= P\left(1+\frac{r}{100}\right)^n \\ &= 800\left(1+\frac{10}{100}\right)^{30} \\ &= 800(1{\cdot}1)^{30} \\ &= \$13\,959{\cdot}52 \\ &= \$13\,960 \text{ (to nearest dollar)} \end{aligned}$$

(ii) Let A_n be the amount to which the investment has grown after 30 years.

After 30 years, $A_{30} = 800(1{\cdot}1)^{30}$

after 29 years, $A_{29} = 800(1{\cdot}1)^{29}$

after 28 years, $A_{28} = 800(1{\cdot}1)^{28}$

$\vdots \qquad \vdots$

after 1 year, $A_1 = 800(1{\cdot}1)^1$

The total investment is:

$$\begin{aligned} A &= A_{30}+A_{29}+A_{28}+\cdots+A_1 \\ &= 800(1{\cdot}1)^{30}+800(1{\cdot}1)^{29}+800(1{\cdot}1)^{28}+\cdots+800(1{\cdot}1)^1 \\ &= 800\left(1{\cdot}1^{30}+1{\cdot}1^{29}+1{\cdot}1^{28}+\cdots+1{\cdot}1\right) \\ &= 800\underbrace{\left(1{\cdot}1^1+1{\cdot}1^2+\cdots+1{\cdot}1^{30}\right)}_{\text{a GP with } S_n=\frac{a(r^n-1)}{r-1}} \\ &= 800\left[\frac{1{\cdot}1\left(1{\cdot}1^{30}-1\right)}{1{\cdot}1-1}\right] \\ &= 800\left[\frac{1{\cdot}1\left(1{\cdot}1^{30}-1\right)}{0{\cdot}1}\right] \\ &= \$144\,754{\cdot}74 \\ &= \$144\,755 \text{ (to nearest dollar)} \end{aligned}$$

Perform this step until you can get it correct without looking at the solution.

19 Let A be the number of insects after 100 days.

This problem is similar to compound interest, with the amount A representing the number of insects instead of money.

(i) $A = P\left(1+\frac{r}{100}\right)^n$, where $P = 1000$, $r = 1$, $n = 100$

$$\begin{aligned} &= 1000\left(1+\frac{1}{100}\right)^{100} \\ &= 1000(1{\cdot}01)^{100} \\ &= 2704{\cdot}8 \\ &\approx 2705 \text{ (to the nearest whole number)} \end{aligned}$$

(ii) Let A_n be the number of insects after n days.

After 200 days, $A_{200} = 1000(1{\cdot}01)^{200}$

after 199 days, $A_{199} = 1000(1{\cdot}01)^{199}$

after 198 days, $A_{198} = 1000(1{\cdot}01)^{198}$

$\vdots \qquad \vdots$

after 1 day, $A_1 = 1000(1{\cdot}01)^1$

The total number is:

$$\begin{aligned} A &= A_1+A_2+A_3+\cdots+A_{200} \\ &= 1000(1{\cdot}01)^1+1000(1{\cdot}01)^2+1000(1{\cdot}01)^3+\cdots+1000(1{\cdot}01)^{200} \end{aligned}$$

$$= 1000\underbrace{\left(1{\cdot}01 + 1{\cdot}01^2 + \cdots + 1{\cdot}01^{200}\right)}_{\text{a GP with } S_n = \frac{a\left(r^n - 1\right)}{r-1}}$$

$$= 1000\left[\frac{1{\cdot}01\left(1{\cdot}01^{200} - 1\right)}{1{\cdot}01 - 1}\right]$$

$$= 1000\left[\frac{1{\cdot}01\left(1{\cdot}01^{200} - 1\right)}{0{\cdot}01}\right]$$

$$= 637\,917{\cdot}803$$

$$= 638\,000 \text{ to the nearest thousand}$$

20 (i) $A = P\left(1 + \frac{r}{100}\right)^n$, where $P = 50\,000$, $r = 1$ (% p.m.)

$$A_1 = 50\,000\left(1 + \frac{1}{100}\right)^1$$

$$= 50\,000(1{\cdot}01)$$

$$= \$50\,500$$

(ii) Let A_1, A_2 be the amounts owing at the end of years 1 and 2.

$$A_1 = P(1{\cdot}01)^{12} - M$$

$$= 50\,000(1{\cdot}01)^{12} - M$$

(iii) $A_2 = A_1(1{\cdot}01)^{12} - M$

$$= \left[50\,000(1{\cdot}01)^{12} - M\right]1{\cdot}01^{12} - M$$

$$= 50\,000(1{\cdot}01)^{24} - \left(1{\cdot}01^{12}\right)M - M$$

$$= 50\,000(1{\cdot}01)^{24} - M\left(1{\cdot}01^{12} + 1\right)$$

When the loan is paid out, $A_2 = 0$

$$\therefore \quad 50\,000(1{\cdot}01)^{24} - M\left(1{\cdot}01^{12} + 1\right) = 0$$

$$M\left(1{\cdot}01^{12} + 1\right) = 50\,000(1{\cdot}01)^{24}$$

$$\therefore M = \frac{50\,000 \times (1{\cdot}01)^{24}}{1{\cdot}01^{12} + 1}$$

(iv) $M = \frac{50\,000 \times (1{\cdot}01)^{24}}{1{\cdot}01^{12} + 1} = \$29\,850{\cdot}47$

Total repaid $= 2M = 2 \times \$29\,850{\cdot}47 = \$59\,700{\cdot}94$

$$\therefore \quad \text{Interest} = \text{total repaid} - \text{principal}$$

$$= 59\,700{\cdot}94 - 50\,000$$

$$= \$9700{\cdot}94$$

21 (i) $A = P\left(1 + \frac{r}{100}\right)^n$, where $P = \$10\,000$, $r = 8$, $n = 10$ years

$$= 10\,000\left(1 + \frac{8}{100}\right)^{10}$$

$$= 10\,000(1{\cdot}08)^{10}$$

$$= \$21\,589{\cdot}25 \text{ (to the nearest cent)}$$

(ii) Let A_n be the amount in the account after n years.

1 Jan 2001: $A_1 = 10\,000(1{\cdot}08)^1 + 1000$

1 Jan 2002: $A_2 = A_1(1{\cdot}08) + 1000$

$$= \left[10\,000(1{\cdot}08)^1 + 1000\right]1{\cdot}08 + 1000$$

(ii) Observe that the question is a combination of a compound-interest problem and a super-annuation problem.

$$= 10\,000(1{\cdot}08)^2 + 1000(1{\cdot}08) + 1000$$

1 Jan 2003: $A_3 = A_2(1{\cdot}08) + 1000$

$$= \left[10\,000(1{\cdot}08)^2 + 1000(1{\cdot}08) + 1000\right]1{\cdot}08 + 1000$$

$$= 10\,000(1{\cdot}08)^3 + 1000(1{\cdot}08)^2 + 1000(1{\cdot}08) + 1000$$

$$\vdots \qquad\qquad \vdots$$

1 Jan 2010: $A_{10} = 10\,000(1{\cdot}08)^{10} + 1000(1{\cdot}08)^9 + 1000(1{\cdot}08)^8 + \cdots + 1000(1{\cdot}08)^1 + 1000$

$$= 10\,000(1{\cdot}08)^{10} + 1000\left(1{\cdot}08^9 + 1{\cdot}08^8 + \cdots + 1{\cdot}08^1 + 1\right)$$

$$= 10\,000(1{\cdot}08)^{10} + 1000\underbrace{\left(1 + 1{\cdot}08 + \cdots + 1{\cdot}08^9\right)}_{\text{GP with } S_n = \frac{a\left(r^n - 1\right)}{r-1}}$$

$$= 10\,000(1{\cdot}08)^{10} + 1000\left[\frac{1\left(1{\cdot}08^{10} - 1\right)}{1{\cdot}08 - 1}\right]$$

$$= 10\,000(1{\cdot}08)^{10} + 1000\left[\frac{1{\cdot}08^{10} - 1}{0{\cdot}08}\right]$$

$$= \$21\,589{\cdot}25 + \$14\,486{\cdot}56$$

$$= \$36\,075{\cdot}81 \text{ (to the nearest cent)}$$

(iii) $A = P\left(1 + \frac{r}{100}\right)$ $\quad\therefore\quad 35\,478 = 10\,000\left(1 + \frac{r}{100}\right)^{10}$

$$3{\cdot}5478 = \left(1 + \frac{r}{100}\right)^{10}$$

$$\therefore \quad 1 + \frac{r}{100} = \sqrt[10]{3{\cdot}5478}$$

$$\frac{r}{100} = 3{\cdot}5478^{\frac{1}{10}} - 1$$

$$\therefore \quad r = 100\left(3{\cdot}5478^{\frac{1}{10}} - 1\right)$$

$$= 13{\cdot}500\ldots$$

i.e. the interest rate is 13·5%.

$\boxed{\text{Alternative method}}$

$35\,478 = 10\,000\left(1 + \frac{r}{100}\right)^{10} \quad \therefore \quad 3{\cdot}5478 = \left(1 + \frac{r}{100}\right)^{10}$

Taking logs of both sides: $\log_{10} 3{\cdot}5478 = \log_{10}\left(1 + \frac{r}{100}\right)^{10}$

$$= 10\log_{10}\left(1 + \frac{r}{100}\right)$$

$$\therefore \quad \log_{10}\left(1 + \frac{r}{100}\right) = \frac{\log_{10} 3{\cdot}5478}{10}$$

$$= 0{\cdot}054\,99\ldots$$

$$\therefore \quad 1 + \frac{r}{100} = 10^{0{\cdot}054\,99\ldots}$$

$$= 1{\cdot}135\,00\ldots$$

$$\therefore \quad \frac{r}{100} = 0{\cdot}135\,00\ldots$$

$$r = 13{\cdot}500\ldots$$

$\therefore$ the annual rate is 13·5%

22 Let A_n be the amount of money in the account at the end of year n

Let $P = \$60\,000$, $r = 7$

$$A = P\left(1+\frac{r}{100}\right)^n = P\left(1+\frac{7}{100}\right)^n = P(1{\cdot}07)^n$$

(i) $A_1' = P(1{\cdot}07) - W = 60\,000(1{\cdot}07) - W$

(ii)
$$\begin{aligned}
A_2 &= A_1(1{\cdot}07) - W \\
&= [P(1{\cdot}07) - W]\,1{\cdot}07 - W \\
&= P(1{\cdot}07)^2 - 1{\cdot}07W - W \\
&= P(1{\cdot}07)^2 - (1{\cdot}07 + 1)W \\
A_3 &= A_2(1{\cdot}07) - W \\
&= \left[P(1{\cdot}07)^2 - (1{\cdot}07 + 1)W\right]1{\cdot}07 - W \\
&= P(1{\cdot}07)^3 - \left(1{\cdot}07^2 + 1{\cdot}07\right)W - W \\
&= P(1{\cdot}07)^3 - \left(1{\cdot}07^2 + 1{\cdot}07 + 1\right)W \\
&\vdots \qquad \vdots \qquad \vdots \\
A_{20} &= P(1{\cdot}07)^{20} - \left(1{\cdot}07^{19} + 1{\cdot}07^{18} + \cdots + 1{\cdot}07 + 1\right)W \\
&= P(1{\cdot}07)^{20} - \underbrace{\left(1 + 1{\cdot}07 + \cdots + 1{\cdot}07^{19}\right)W}_{\text{a GP with } S_n = \frac{a\left(r^n - 1\right)}{r-1}} \\
&= P(1{\cdot}07)^{20} - \left[\frac{1\left(1{\cdot}07^{20} - 1\right)}{1{\cdot}07 - 1}\right]W \\
&= P(1{\cdot}07)^{20} - \left(\frac{1{\cdot}07^{20} - 1}{0{\cdot}07}\right)W \\
&= 60\,000(1{\cdot}07)^{20} - \left(\frac{1{\cdot}07^{20} - 1}{0{\cdot}07}\right)W
\end{aligned}$$

(iii) The account is empty when $A_{20} = 0$

$$\begin{aligned}
\therefore \quad 0 &= 60\,000(1{\cdot}07)^{20} - \left(\frac{1{\cdot}07^{20} - 1}{0{\cdot}07}\right)W \\
W\left(\frac{1{\cdot}07^{20} - 1}{0{\cdot}07}\right) &= 60\,000(1{\cdot}07)^{20} \\
\therefore \quad W &= \frac{60\,000(1{\cdot}07)^{20} \times 0{\cdot}07}{1{\cdot}07^{20} - 1} \\
&= \$5663{\cdot}58 \text{ (to the nearest cent)}
\end{aligned}$$

(iv) Since she wishes to withdraw \$7500 p.a., we must try different values of r to give results close to this amount.

Since $r\% = 0{\cdot}07 \rightarrow 5663 \quad \therefore\ r\% > 0{\cdot}07$

Try $r\% = 0{\cdot}10$: $W = \dfrac{60\,000 \times 1{\cdot}1^{20} \times 0{\cdot}1}{1{\cdot}1^{20} - 1}$

$= \$7047{\cdot}58 \qquad \therefore\ r$ is too low

Try $r\% = 0{\cdot}12$: $W = \dfrac{60\,000 \times 1{\cdot}12^{20} \times 0{\cdot}12}{1{\cdot}12^{20} - 1}$

$= \$8032{\cdot}73 \qquad \therefore\ r$ is too high

Try $r\% = 0{\cdot}11$: $W = \dfrac{60\,000 \times 1{\cdot}11^{20} \times 0{\cdot}11}{1{\cdot}11^{20} - 1}$

$= \$7534{\cdot}54 \qquad \therefore\ r$ is close enough

Thus Glenys would need to obtain 11% p.a. interest to withdraw \$7500 p.a.

A time-payment type of problem.

Use P, until ready to substitute, to save working space and time in writing.